SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	**Assessment:**	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	**Total Marks:**	20 (2 Questions × 10Marks)
CO1 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)*		
Student Name:	P.Moshiya	**Reg. No.:**	192572142
Section / Batch:	2025	**Date:**	

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Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
	GRAND TOTAL:	**20 Marks**	

Annexure A –Case Study

1.A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

2.An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Faculty In-charge
Signature: _______________
Date: ___________________

1. Limitations of Edge Detection and Advantages of Harris Corner Detection

- **Edge detection** identifies boundaries where image intensity changes sharply. It may produce long edges rather than unique points, making accurate image matching difficult.
- Edges can change significantly due to **illumination, rotation, shadows, and background changes**.

- Edge-based methods may generate many similar edge points, causing **incorrect correspondences** between satellite images.
- **Harris Corner Detection** identifies corners, where intensity changes strongly in two directions.
- Corners are more **distinctive and stable feature points**, so they can be matched more reliably between images.
- Harris corners provide better **feature localization**, improving the estimation of translation, rotation, and alignment.
- Therefore, Harris Corner Detection gives **higher registration accuracy and better robustness** under varying imaging conditions.

2. Color, Grayscale and Binary Representations

- **Color images** contain RGB information and provide more details, but they require more memory and computation. This can slow down real-time object detection.
- **Grayscale images** contain only intensity information, reducing data size and computational complexity. Edges and shapes can still be detected effectively.
- **Binary images** contain only two values (0 and 1). They are computationally efficient but lose many details, which can reduce the accuracy of detecting objects with complex shapes or similar backgrounds.
- For **edge detection and feature extraction**, grayscale generally provides a good balance between useful information and processing speed.
- Therefore, **grayscale representation is the most appropriate choice** for real-time surveillance. It reduces computational cost while preserving important shape, edge, and texture information needed for reliable vehicle and pedestrian detection.